

Projective Information Loss and Fibre Erasure in Admissible Non-Injective Projection

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Abstract

We establish a fibre-erasure theorem for the admissible coarse-graining hierarchy of the Cosmochrony spectral programme [3, 1]. The non-injective projection $\Pi : \Omega \rightarrow \mathcal{O}$ defines a residual fibre-information functional $I(c; \sigma(\ell))$, where c is a Weil-block fibre label and $\sigma_c(\ell)$ is the BFS capacity profile at coarse-graining depth ℓ . Conditional on the structural sufficiency hypothesis [H-suff] — that $\sigma(\ell)$ is a sufficient statistic for the hierarchy beyond ℓ , a condition numerically supported for $q \in \{61, 151, 211\}$ (Section 5.3) — the data-processing inequality [9] implies that $I(c; \sigma(\ell))$ is non-increasing in ℓ : fibre information is erased, not created, under admissible coarse-graining. This result is *not* a c -theorem; it does not count effective degrees of freedom. It is a quantitative realisation of the ENI no-go theorem [3]: non-injectivity of Π forces projective information loss, and the data-processing inequality makes this loss monotone and measurable. The inter-sector variance $\text{Var}_c(\sigma_c(\ell))$ serves as a computable proxy; its restriction to $\text{SU}(3)$ colour-triplets connects directly to hypothesis [H-color] [8, 7].

Keywords. non-injective projection; fibre erasure; admissible coarse-graining; data-processing inequality; information monotone; BFS capacity; spectral admissibility; projective information loss; Weil representation; Heisenberg group; [H-color]

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1 Introduction

The ENI no-go theorem [3] establishes a fundamental structural fact: genuine emergence is necessarily non-injective. Any surjective map $\Pi : \Omega \rightarrow \mathcal{O}$ satisfying the projective-emergent conditions carries a strictly positive projection entropy $S_\Pi > 0$: information about the microscopic representative is irrecoverably lost at the effective level. This is a qualitative result. It identifies the structure of information loss but says nothing about how this loss is distributed across a hierarchy of resolutions.

The Cosmochrony spectral programme [5, 6] provides a concrete realisation of this hierarchy. For each prime q , the BFS capacity profiles $\sigma_c(\ell)$, indexed by Weil-block fibre label c and coarse-graining depth ℓ , track the residual projective capacity of block c as the BFS cascade advances. Numerical campaigns across $q \in \{61, 151, 211, 307\}$ reveal a consistent pattern: inter-sector variance collapses as depth increases, and the profiles converge toward a universal profile σ^* approached in the limit $q \rightarrow \infty$ (Section 5.4). Hypothesis [H-color] [8, 7] identifies one specific instance of this convergence: the $SU(3)$ colour-triplet sectors $\{c, \omega c, \omega^2 c\}$ become indistinguishable at the effective level for $q \equiv 1 \pmod{3}$.

This paper addresses the following question: can the qualitative ENI no-go and the quantitative spectral convergence be connected by a principled information-theoretic structure? We show that the answer is yes, conditional on the structural sufficiency hypothesis [H-suff]: the residual fibre-information functional $I(c; \sigma(\ell))$ is non-increasing in the coarse-graining depth ℓ , by the data-processing inequality [9]. The result is conditional, not unconditional: [H-suff] is strongly motivated (Remark 3.2), numerically supported for $q \in \{61, 151, 211\}$ (Section 5.3), but not yet proved analytically.

Two structural distinctions are essential before proceeding. First, the coarse-graining axis ℓ is distinct from and orthogonal to the temporal cascade axis n of the PTO paper [4]: see Table 1 and Remark 2.3. Second, as stated in Section 1.1, ENI alone does not imply monotonicity; the hierarchical structure of [H-suff] and [H-sg] is the informational content this paper contributes beyond ENI.

1.1 Contributions and non-claims

This paper establishes, conditional on hypotheses [H-suff] and [H-sg] (Section 2.4):

- (i) a monotone information functional $I(c; \sigma(\ell))$ for the BFS coarse-graining hierarchy (Theorem 4.3);
- (ii) its non-increase under depth-increment, via the data-processing inequality;
- (iii) the identification of $I(c; \sigma(\ell)) = 0$ as the fibre-erasure equilibrium condition (Corollary 4.4).

This paper does *not* establish:

- (i) a c -theorem or analogue (no counting of effective degrees of freedom; see open problem [O-2]);
- (ii) hypothesis [H-suff] itself (stated and motivated; numerically supported for $q \in \{61, 151, 211\}$; analytical proof remains open, see [O-1]);
- (iii) the semigroup hypothesis [H-sg] (stated and motivated; see open problem [O-1]).

Strict disclaimer. This result should not be called a c -theorem. It is a *fibre-erasure theorem*: the monotone measures the loss of residual fibre-label information under admissible coarse-graining. A c -theorem-type statement would require an additional counting principle for effective degrees of freedom, which is not established here and is listed as open problem [O-2] in Section 6.

ENI versus monotonicity. ENI [3] establishes that information is structurally lost under non-injective projection, but does not by itself imply any monotone hierarchy over the coarse-graining scale ℓ . The monotonicity of Theorem 4.3 requires the additional hierarchical structure encoded in [H-suff] and [H-sg]: these are the informational content this paper contributes beyond ENI.

1.2 Programme placement

The present paper occupies a hub position in the programme. ENI [3] and Foundation [1] provide the foundational input: the non-injective projection Π , the Weil-fibre realisation, and the projection entropy S_Π . The spectral admissibility programme [5, 6] provides the operational hierarchy: the BFS capacity profiles $\sigma_c(\ell)$ and their numerical campaigns, including the O31–O32 investigation [8, 7]. This paper connects these two branches by translating the ENI projection entropy into the operational functional $I(c; \sigma(\ell))$ and identifying spectral convergence $\sigma_c \rightarrow \sigma^*$ as the thermodynamic universal profile (Section 5.4). The PTO paper [4] handles the complementary temporal axis n (Table 1); the two papers are structurally orthogonal and together constitute the information-theoretic architecture of the admissibility cascade.

2 Framework

2.1 Non-injective projection and fibre structure

We recall the setting of [3, 1].

Definition 2.1 (Non-injective projection and fibre). A *projective description* is a surjective map $\Pi : \Omega \rightarrow \mathcal{O}$ such that $\mathcal{O}$ carries no microstate-resolving structure independent of Π . The *fibre* over $o \in \mathcal{O}$ is $\Pi^{-1}(o) = \{\omega \in \Omega : \Pi(\omega) = o\}$. The *projection entropy* is $S_\Pi = \mathbb{E}_o[\log |\Pi^{-1}(o)|] \geq 0$, with $S_\Pi > 0$ whenever Π is non-injective (ENI Theorem 2.1, [3]).

In the spectral realisation [1], the fibre structure is implemented via the Weil representation V_ρ of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for a prime q . The minimal non-injectivity is the parity involution $c \leftrightarrow q - c$: Weil blocks ρ_c and ρ_{q-c} map to the same effective observable class. The *fibre label* $c \in \{1, \dots, (q-1)/2\}$ indexes blocks within an equivalence class under Π .

2.2 The BFS coarse-graining hierarchy

Definition 2.2 (BFS capacity profile). For a prime q , a fibre label c , and BFS depth $\ell \in \mathbb{N}$, the *BFS capacity profile* $\sigma_c(\ell) \in [0, 1]$ is the normalised residual projective capacity of block c at depth ℓ along the Heisenberg Cayley graph BFS cascade [5, 6]. The *pair observable* is $\sigma_{\text{pair}}(\ell) = \sigma_c(\ell) \cdot \sigma_{q-c}(\ell)$ on conjugate pair $(c, q - c)$.

For a prime q , the Heisenberg group $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ carries a Cayley graph whose BFS shells B_ℓ grow as $|B_\ell| \sim \ell^4$ in the Heisenberg regime [5]. The capacity profile $\sigma_c(\ell)$ measures the residual Gram-Schmidt span of Weil block ρ_c not yet exhausted at shell depth ℓ : it starts at $\sigma_c(0) = 1$ and decreases toward 0 as BFS saturation is approached. The depth parameter $\ell \in \mathbb{N}$ is discrete; the coarse-graining semigroup of Section 2.4 therefore acts on a discrete monoid, not a continuous scale.

2.3 Three-axis separation

Remark 2.3 (Cascade and coarse-graining are orthogonal). The admissible cascade $U_0 \rightarrow U_1 \rightarrow \dots$ defines the temporal axis n and is *not* an RG flow. RG-like behaviour is carried by Π and the semigroup $\{G_{\ell \leftarrow \ell}\}$ on the ℓ -axis. The cascade *enlarges* the admissible space as temporal ordering advances; an RG flow toward the IR *reduces* effective degrees of freedom. These are distinct

Axis	Parameter	Role
Coarse-graining / resolution	ℓ (BFS depth/window)	coarse-graining hierarchy; carries $I(c; \sigma(\ell))$; this paper
Thermodynamic / continuum	$q \rightarrow \infty$	Limit for σ^* ; established numerically (Section 5.4)
Temporal / cascade	n (cascade step)	Projective time; carries $I(n)$ of [4]

Table 1: The three distinct axes of the observable hierarchy. The coarse-graining axis ℓ is the support of this paper. The temporal axis n is the support of PTO [4]. The continuum axis $q \rightarrow \infty$ is the thermodynamic limit in which the fixed point σ^* is approached; Section 5.4 establishes this numerically.

phenomena on orthogonal axes. The fibre-erasure fixed point σ^* lives on the thermodynamic axis: it is approached as $q \rightarrow \infty$ with $O(q^{-1/2})$ corrections, not as a Wilsonian ℓ -limit at fixed q (Section 5.4).

2.4 The coarse-graining semigroup

Definition 2.4 (Coarse-graining channel). For $\ell' > \ell$, the *coarse-graining channel* $G_{\ell' \leftarrow \ell} : \sigma(\ell) \mapsto \sigma(\ell')$ is the map sending the BFS capacity profile at depth ℓ to the profile at depth ℓ' .

Hypothesis 2.5 ([H-sg]: semigroup property). The coarse-graining channels satisfy

$$G_{\ell_3 \leftarrow \ell_2} \circ G_{\ell_2 \leftarrow \ell_1} = G_{\ell_3 \leftarrow \ell_1} \quad \text{for all } \ell_1 < \ell_2 < \ell_3. \quad (1)$$

Hypothesis [H-sg] is plausible from the nested structure of BFS shells: the subgraph induced by B_{ℓ_1} is contained in that induced by B_{ℓ_2} for $\ell_1 < \ell_2$, so the transition $\sigma(\ell_1) \rightarrow \sigma(\ell_2)$ can in principle be factored through $\sigma(\ell_1)$ alone. Without [H-sg], the data-processing inequality applied step-by-step via [H-suff] yields $I(c; \sigma(\ell+1)) \leq I(c; \sigma(\ell))$ for each consecutive pair, but not the uniform statement of Theorem 4.3 for all $\ell' > \ell$. Note that the semigroup acts on the space of observable descriptions, not on the static substrate Ω : composability is a property of the coarse-graining of effective observables.

2.5 Probability measure on fibre labels

Definition 2.6 (Fibre-label measure). Let μ be a probability measure on $\{c : 1 \leq c \leq (q-1)/2\}$. Two canonical choices are used:

- (i) *Uniform*: $\mu(c) = 2/(q-1)$ for all c .
- (ii) *Triplet-uniform* (for $q \equiv 1 \pmod{3}$): μ uniform on $\{c, \omega c, \omega^2 c\}$, where ω is a primitive cube root of unity modulo q ; used in the [H-color] context [8, 7].

The choice of μ does not affect the monotonicity of $I(c; \sigma(\ell))$ (which holds for any fixed measure by DPI), but affects its value and the interpretation of the fixed point.

3 Structural Sufficiency and the Markov Property

Hypothesis 3.1 ([H-suff]: structural sufficiency). For all $\ell' > \ell$, the coarse-graining channel $G_{\ell' \leftarrow \ell}$ is conditionally independent of the fibre label c given $\sigma(\ell)$:

$$c \longrightarrow \sigma(\ell) \longrightarrow \sigma(\ell'). \quad (2)$$

That is, $\sigma(\ell)$ is a *sufficient statistic* for $\sigma(\ell')$ with respect to the fibre label c . The term “sufficient” is used in the information-theoretic sense: all information in $\sigma(\ell)$ that is predictive of $\sigma(\ell')$ is already fully encoded; no additional knowledge of c reduces the uncertainty in $\sigma(\ell')$ beyond what $\sigma(\ell)$ already captures.

Remark 3.2 (Interpretive status of [H-suff]). Hypothesis [H-suff] is not introduced solely to enable the DPI. It is the informational reformulation of an effective closure property already implicit throughout the spectral admissibility programme: if [H-suff] failed — if BFS dynamics at depth ℓ' re-accessed label information not encoded in $\sigma(\ell)$ — then $\sigma_c(\ell)$ would not constitute stable effective observables under Π , and the convergence $\sigma_c \rightarrow \sigma^*$, the plateau structure across $q \in \{61, 151, 211, 307\}$, and the [H-color] campaign [8, 7] would be structurally very difficult to account for. More precisely, [H-suff] is the condition that the BFS observable $\sigma(\ell)$ constitutes a *complete summary* of the fibre structure at resolution ℓ : the projection Π has already collapsed the relevant fibre information at each level of the hierarchy, and no deeper BFS step retrieves it. This is exactly what the non-injectivity of Π (ENI [3]) demands at the effective level: fibre information is not gradually re-injected as coarse-graining proceeds, but is collapsed once and for all by Π . In this sense, [H-suff] makes explicit, in information-theoretic language, a structural commitment already implicit in the programme's use of $\sigma_c(\ell)$ as an effective observable.

Remark 3.3 (Status of [H-suff]). Hypothesis [H-suff] is strongly motivated by Remark 3.2 and is compatible with all known spectral results. A permutation-based Spearman test on BFS capacity data for $q \in \{61, 151, 211\}$ yields no significant c -dependence in step-by-step residuals at any tested depth (Section 5.3), providing numerical evidence for [H-suff]. Analytical proof remains open; see open problem [O-1].

Remark 3.4 (Why [H-suff] is load-bearing). If hypothesis [H-suff] fails for some pair (ℓ, ℓ') — that is, if $\sigma(\ell')$ depends on c beyond its dependence on $\sigma(\ell)$ — then no structural information monotone of the form $I(c; \sigma(\ell')) \leq I(c; \sigma(\ell))$ can be guaranteed. If $p(\sigma(\ell') \mid \sigma(\ell), c) \neq p(\sigma(\ell') \mid \sigma(\ell))$ for some c , then the chain $c \rightarrow \sigma(\ell) \rightarrow \sigma(\ell')$ is not Markov, and the data-processing inequality does not apply. This is not a limitation of the proof technique; it reflects the genuine logical dependence of the monotonicity on the Markov structure.

A proof of [H-suff] would establish that the BFS depth-increment kernel $p(\sigma(\ell+1) \mid \sigma(\ell), c)$ is independent of the fibre label c . Structurally, this would follow from the fact that the BFS recursion on the Heisenberg Cayley graph is governed by the group geometry — shared across all fibre labels up to conjugation — and the Born-Infeld admissibility constraint, rather than by the specific character of block c . A partial result is already in hand: the conjugation symmetry $\rho_c \cong \rho_{q-c}$ ensures $\sigma_c(\ell) = \sigma_{q-c}(\ell)$ within each conjugate pair, consistent with [H-suff] at the pair level. The full extension requires showing that cross-pair dynamics also factors through $\sigma(\ell)$.

The full extent of [H-suff] — including cross-pair dynamics — has been tested numerically for $q \in \{61, 151, 211\}$; see Section 5.3.

4 The Fibre-Erasure Theorem

4.1 The information functional

Definition 4.1 (Residual fibre information). The *residual fibre information* at depth ℓ is

$$I(c; \sigma(\ell)) = H(c) - H(c \mid \sigma(\ell)) = H(\sigma(\ell)) - H(\sigma(\ell) \mid c), \quad (3)$$

where $H(\cdot)$ denotes Shannon entropy with respect to measure μ .

Remark 4.2 (Physical interpretation). $I(c; \sigma(\ell)) = 0$ if and only if $\sigma(\ell) \perp c$: the BFS profile carries no information about the microscopic fibre representative. This is the operational statement of fibre erasure under Π .

4.2 Main theorem

Theorem 4.3 (Fibre-Erasure Theorem). *Assume hypotheses [H-suff] (Hypothesis 3.1) and [H-sg] (Hypothesis 2.5). Then for all $\ell' > \ell \geq 0$:*

$$I(c; \sigma(\ell')) \leq I(c; \sigma(\ell)). \quad (4)$$

In particular, $I(c; \sigma(\ell))$ is non-increasing in ℓ and converges to a limit $I_\infty \geq 0$.

Proof sketch. By [H-suff], the triple $(c, \sigma(\ell), \sigma(\ell'))$ forms a Markov chain (2). The data-processing inequality [9] applied to this chain gives (4) directly. Composability [H-sg] extends the inequality from $\ell' = \ell + 1$ to all $\ell' > \ell$. $\square$

Corollary 4.4 (Fibre-erasure equilibrium condition). *Under the hypotheses of Theorem 4.3, $I(c; \sigma(\ell)) = 0$ if and only if $\sigma(\ell) \perp c$. This is the fibre-erasure equilibrium: the coarse-graining has fully collapsed the fibre-label information. It is realised by the thermodynamic universal profile σ^* in the limit $q \rightarrow \infty$ (Remark 5.3).*

Explicit disclaimer. Theorem 4.3 is *not* a c -theorem. At the fibre-erasure equilibrium, $I_\infty = 0$ does not count any effective degrees of freedom. A c -theorem analog requires a separate counting functional; see open problem [O-2].

4.3 Information decomposition and programme connections

By the chain rule for entropy:

$$H(\sigma(\ell)) = H(\sigma(\ell) \mid c) + I(c; \sigma(\ell)). \quad (5)$$

Here $I(c; \sigma(\ell))$ is the fibre-erasure functional (Theorem 4.3), monotone non-increasing in ℓ ; $H(\sigma(\ell) \mid c)$ is the intrinsic entropy of the observable conditioned on the fibre label, and is the natural candidate for a counting monotone [O-2].

Remark 4.5 (Connection to PTO and axis orthogonality). The temporal monotone $I(n) = \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n)$ of [4] lives on the temporal axis n (Table 1). The present $I(c; \sigma(\ell))$ lives on the coarse-graining axis ℓ . They measure distinct phenomena on orthogonal axes and should not be conflated.

5 Numerical Proxies and Spectral Programme Connections

5.1 Inter-sector variance as proxy

Definition 5.1 (Inter-sector variance). The *inter-sector variance* at depth ℓ is

$$\mathcal{V}(\ell) = \text{Var}_c(\sigma_c(\ell)) = \mathbb{E}_c[\sigma_c(\ell)^2] - (\mathbb{E}_c[\sigma_c(\ell)])^2. \quad (6)$$

$\mathcal{V}(\ell) = 0$ is a necessary condition for $I(c; \sigma(\ell)) = 0$, but $\mathcal{V}$ is *not* guaranteed to be monotone in ℓ . Non-monotone behaviour of $\mathcal{V}$ would not contradict Theorem 4.3; it would indicate that variance is an imperfect surrogate for $I(c; \sigma(\ell))$.

5.2 Triplet variance and [H-color]

For $q \equiv 1 \pmod{3}$, the *triplet-restricted variance* is

$$\mathcal{V}_{\text{triplet}}(\ell) = \text{Var}(\sigma_c(\ell), \sigma_{\omega c}(\ell), \sigma_{\omega^2 c}(\ell)), \quad (7)$$

averaged over colour triplets.

Remark 5.2 ([H-color] as a fibre-erasure statement). Hypothesis [H-color] of [8, 7] asserts $\sigma_c(\ell) = \sigma_{\omega c}(\ell) = \sigma_{\omega^2 c}(\ell)$ for $q \equiv 1 \pmod{3}$, which is precisely $\mathcal{V}_{\text{triplet}}(\ell) = 0$ and, under [H-suff], $I_{\text{triplet}}(c; \sigma(\ell)) = 0$. The fibre-erasure theorem provides the information-theoretic framework within which [H-color] is a special case: the colour label becomes unobservable at the effective level. The triplet-restricted variance is already computable from the $q \in \{61, 151, 211, 307\}$ runs of O32 [7].

5.3 Numerical evidence for [H-suff]

Hypothesis [H-suff] was tested numerically on the BFS capacity data from the O25 pipeline [5, 6] for $q \in \{61, 151, 211\}$. For each depth step $\ell \rightarrow \ell + 1$, the k -nearest-neighbour regression residuals $e_c = \sigma_c(\ell + 1) - \hat{f}(\sigma_c(\ell))$ are computed from all $q - 1$ block-averaged mean trajectories, where $\hat{f}$ is estimated without reference to c . The Spearman correlation $|\rho(e_c, c)|$ is then computed and compared to a permutation null (500 shuffles of c), yielding a right-tail p -value and a standardised z -score. Under [H-suff], the residuals e_c should be independent of c , and $|\rho|$ should be statistically indistinguishable from zero at every depth.

q	depths tested	$ \rho _{\max}$	z -score max	signif. ($p < 0.05$)
61	10	0.169	0.78	0/10
151	37	0.123	1.05	0/37
211	49	0.120	1.55	0/49

Table 2: Spearman test of conditional independence $\sigma(\ell + 1) \perp c \mid \sigma(\ell)$ on BFS capacity data for $q \in \{61, 151, 211\}$. $|\rho|_{\max}$: maximum absolute Spearman correlation between residuals and the fibre label over all tested depth steps. z -score: standardised deviation from the permutation null. Significant: number of depth steps with $p < 0.05$; zero in all cases.

For all three primes and across all depth steps tested, $|\rho(e_c, c)|$ does not exceed 0.17 and no depth step yields $p < 0.05$. The results, shown in Figure 1, are consistent with the permutation null throughout: no systematic c -dependence in the step-by-step residuals is detected.

These results constitute numerical evidence that the BFS depth-increment dynamics does not depend on the fibre label c beyond the capacity profile $\sigma(\ell)$, consistent with [H-suff].

Limitations. The test is limited by the available data: the O25 pipeline [5, 6] stores only the mean BFS capacity profile averaged over M block samples per fibre label; per-block trajectories would enable a more powerful categorical test. No full O25 run is currently available for $q = 307$, though [H-color] at $q = 307$ has been confirmed directly by O32 [7].

[H-color] co-admissibility diagnostic. The leave-triplet-out η^2 test (panel C of Figure 1) detects strong within-triplet residual correlation at $q = 151$ ($z_{\max} = 3.07$) and moderately at $q = 211$ ($z_{\max} = 3.25$), consistent with the [H-color] co-admissibility established by O32 [7]: triplet members share equal σ profiles, giving equal residuals after leave-triplet-out regression. This is not evidence against [H-suff]; it is an incidental corroboration of [H-color]. The signal increases with better [H-color] quality: $q = 151$ has the smallest inter-triplet variance ratio ($R_{\text{var}} = 0.71 \times 10^{-3}$) of the tested primes, consistent with the strongest triplet signal.

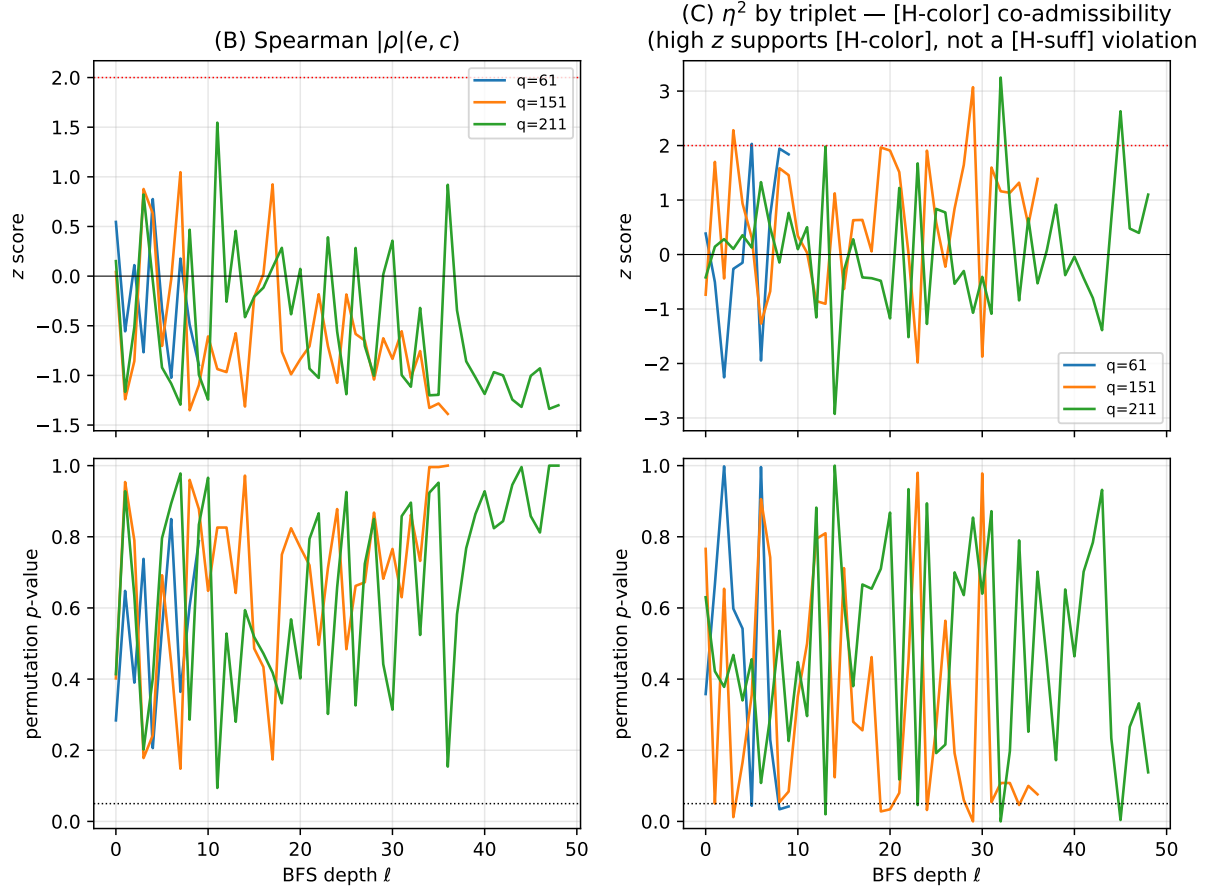


Figure 1: Numerical test of [H-suff] for $q \in \{61, 151, 211\}$. **Left panels (B):** Spearman $|\rho|(e, c)$ test (z -scores and p -values vs BFS depth ℓ). All z -scores lie below the $z = 2$ threshold (dashed red); no depth is significant. **Right panels (C):** leave-triplet-out η^2 test grouped by SU(3) colour triplet for $q \equiv 1 \pmod{3}$; this measures [H-color] co-admissibility quality and does not test [H-suff] directly (see text).

5.4 Axis disambiguation: the fixed point is thermodynamic

The question of whether the fibre-erasure fixed point σ^* — the configuration where $I(c; \sigma(\ell)) = 0$ and fibre-label information is fully collapsed — is approached as a limit in the coarse-graining depth ℓ at fixed q (a Wilsonian fixed point of the semigroup $\{G_{\ell' \leftarrow \ell}\}$) or only in the thermodynamic limit $q \rightarrow \infty$ was investigated numerically on the BFS capacity data for $q \in \{61, 151, 211\}$. Two complementary diagnostics were computed.

Coefficient of variation. The scale-free inter-sector spread

$$\text{CV}(\ell) = \frac{\text{std}_c(\sigma_c(\ell))}{\text{mean}_c(\sigma_c(\ell))} \quad (8)$$

equals $\text{std}(A_c)/\text{mean}(A_c)$, a depth-independent constant, if and only if profiles differ purely by amplitude: $\sigma_c(\ell) = A_c \sigma^*(\ell)$. Decrease of $\text{CV}(\ell)$ to zero would indicate convergence to a common profile, i.e. a Wilsonian fixed point at fixed q . The data show $\text{CV}(\ell)$ starting small at the fitting-window entry (0.045–0.079 across $q \in \{61, 151, 211\}$) and growing throughout the pre-saturation regime at all three primes (Figure 2, left panels); no plateau or decrease toward zero is observed. This rules out Wilsonian convergence at fixed q .

Amplitude-ratio test. For 20 random pairs (c, c') , the coefficient of variation of the ratio $\sigma_c(\ell)/\sigma_{c'}(\ell)$ over depth measures shape differences beyond amplitude scaling. Table 3 shows this quantity decreasing with q . The trend from $q = 61$ (0.156) to $q \in \{151, 211\}$ (0.065–0.062) is

q	ratio-CV mean	ratio-CV std
61	0.156	0.069
151	0.065	0.021
211	0.062	0.025

Table 3: Amplitude-ratio CV over 20 random pairs (c, c') : coefficient of variation of $\sigma_c(\ell)/\sigma_{c'}(\ell)$ over depth in the pre-saturation regime. The decrease with q is consistent with $O(q^{-1/2})$ shape corrections from U1 universality [5, 6].

consistent with $O(q^{-1/2})$ corrections from U1 universality [5, 6]: at larger q , profiles approach a common shape.

Cross- q mean profile comparison. After normalising each mean profile by its value at the fitting-window start $\ell_0 = n_0$, the profiles for $q = 151$ and $q = 211$ are nearly identical, while $q = 61$ decays more rapidly (Figure 2, right panel). No collapse onto a single q -universal curve is observed at finite q .

These results establish that σ^* is a thermodynamic limit object: it is approached as $q \rightarrow \infty$ with $O(q^{-1/2})$ amplitude corrections from U1 universality, not as a Wilsonian ℓ -limit at fixed q . At finite q , approximate fixed points $\sigma^*(q)$ exist with c -dependent amplitude corrections A_c that vanish as $q \rightarrow \infty$; the universal fixed point requires the continuum limit.

Remark 5.3 (Residual fibre information at finite q ; terminology). Theorem 4.3 and Corollary 4.4 hold for any fixed q : their conclusion is that $I(c; \sigma(\ell))$ is non-increasing; they do not require the equilibrium condition to be reached. At finite q , a residual $I(c; \sigma(\ell)) > 0$ persists at all depths, reflecting the c -dependent amplitude corrections A_c .

A terminological precision is warranted. The *fibre-erasure fixed point* of Corollary 4.4 is the equilibrium condition $I(c; \sigma) = 0$ on the information functional — not a fixed point of the semigroup $\{G_{\ell' \leftarrow \ell}\}$, which has no non-trivial fixed point at finite q (all profiles eventually reach $\sigma_c(\ell) \rightarrow 0$ by BFS saturation). The universal profile σ^* realising this equilibrium condition is

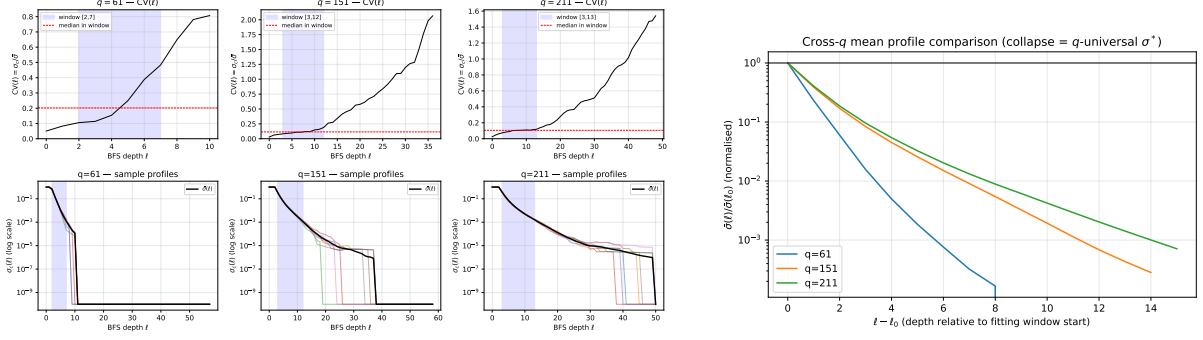


Figure 2: **Left:** $CV(\ell)$ (top row) and sample profiles on a log scale (bottom row) for $q \in \{61, 151, 211\}$. Blue band: fitting window $[n_0, n_1]$. $CV(\ell)$ grows monotonically throughout the pre-saturation regime; the sharp rise beyond the window is a saturation artefact (different c values reach $\sigma_c \rightarrow 0$ at different depths). No plateau or decrease toward zero is observed. **Right:** cross- q mean profiles normalised at $\ell_0 = n_0$. The $q = 151$ and $q = 211$ profiles are nearly identical; $q = 61$ decays faster. The absence of collapse confirms that σ^* requires $q \rightarrow \infty$.

therefore better understood as the *thermodynamic universal profile*: $\sigma^* = \lim_{q \rightarrow \infty} \sigma^*(q)$, where $\sigma^*(q)$ are approximate equilibria at finite q with $I(c; \sigma^*(q)) > 0$, approaching exact equilibrium only in the projective continuum limit $q \rightarrow \infty$ (Conjecture 4.7 of [2]).

5.5 Numerical evidence for [O-2]: conditional entropy monotonicity

The decomposition (5) identifies $H(\sigma(\ell) \mid c)$ as a candidate counting monotone (open problem [O-2]). A numerical proxy is available from the O25 pipeline extended with squared-profile accumulators: for each pair $(c, q - c)$ and BFS depth ℓ , the Bessel-corrected intra-block variance

$$\text{Var}_{\text{intra}}(\ell, c) = \frac{M}{M-1} \left(\mathbb{E}_{\text{block}}[\sigma_c(\ell)^2] - \mathbb{E}_{\text{block}}[\sigma_c(\ell)]^2 \right) \quad (9)$$

estimates the spread of $\sigma_c(\ell)$ across the M independent block samples for fixed fibre label c . Under a Gaussian approximation, $H(\sigma(\ell) \mid c) \approx \text{mean}_c \left[\frac{1}{2} \log(2\pi e \text{Var}_{\text{intra}}(\ell, c)) \right]$, and its monotonicity reduces to that of the proxy

$$h_{\text{intra}}(\ell) = \text{mean}_c \left[\frac{1}{2} \log \text{Var}_{\text{intra}}(\ell, c) \right]. \quad (10)$$

Result for $q = 61$. On the BFS capacity data for $q = 61$ (30 conjugate pairs, $M = 20$ block samples, fitting window $[2, 7]$), $h_{\text{intra}}(\ell)$ is perfectly monotone non-increasing over the pre-saturation regime: zero violations over five consecutive depth increments, linear-trend t -statistic -25.5 , and a reduction of the intra-block standard deviation by a factor 0.024 (approximately $41\times$) over the window. Figure 3 shows $h_{\text{intra}}(\ell)$, the between- c entropy proxy $h_{\text{inter}}(\ell)$, and the normalised spread $\text{CV}_{\text{intra}}(\ell)$.

The normalised spread $\text{CV}_{\text{intra}}(\ell) = \text{mean}_c [\text{std}_{\text{intra}} / \bar{\sigma}_c]$ increases over the same range ($0.46 \rightarrow 2.05$). This reflects the staggered-saturation artefact identified in Section 5.4: at late depths, some of the $M = 20$ blocks reach $\sigma_c = 0$ while others have not, inflating relative within- c variance. It is not a violation of [O-2]; the absolute proxy h_{intra} is the correct test statistic.

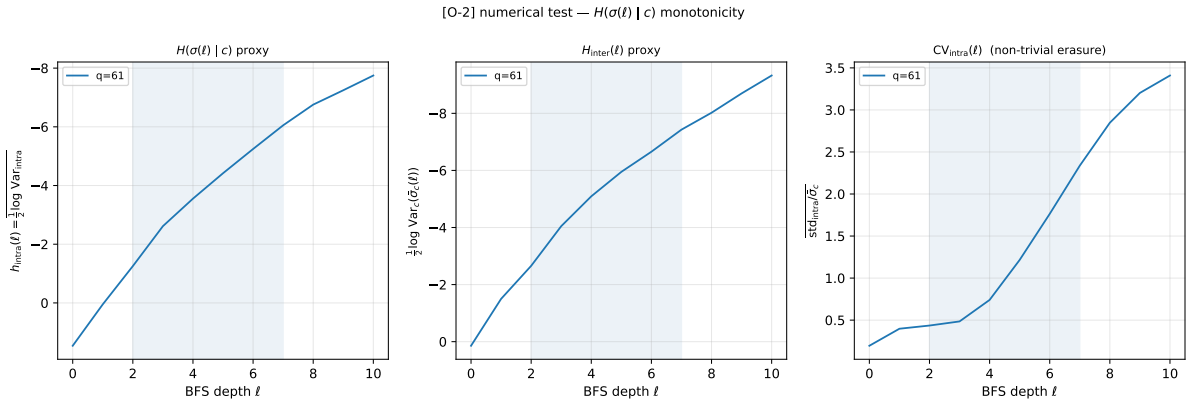


Figure 3: Preliminary numerical test of [O-2] for $q = 61$. **Left:** $h_{\text{intra}}(\ell)$, proxy for $H(\sigma(\ell) \mid c)$; perfectly monotone non-increasing over the fitting window (blue band), zero violations, $t = -25.5$. **Centre:** $h_{\text{inter}}(\ell)$, between- c entropy proxy. **Right:** $\text{CV}_{\text{intra}}(\ell)$; the increase at late depths is the staggered-saturation artefact (Section 5.4), not a violation of [O-2].

These results constitute preliminary numerical evidence that $H(\sigma(\ell) \mid c)$ is monotone non-increasing, consistent with [O-2].

Status of the Gaussian proxy. The quantity $h_{\text{intra}}(\ell)$ is not an exact Shannon entropy unless the within- c distribution of $\sigma_c(\ell)$ is Gaussian. It should be understood as a variance-based entropy proxy. Its monotonicity is nevertheless meaningful: any unimodal within- c distribution whose scale concentrates with ℓ has decreasing entropy under standard scale-sensitive estimators. Thus the present test probes concentration of the intrinsic observable spread, not a full non-parametric entropy estimate.

Counting requirement. A genuine counting monotone requires more than monotonicity alone. One must also show that $H(\sigma(\ell) \mid c)$ approaches a non-zero, stable, and interpretable value at the approximate equilibrium $\sigma^*(q)$, with a controlled q -dependence. If this limiting value vanished identically, the functional would not count effective degrees of freedom in the sense required by a c -theorem analog.

Role of the multi- q extension. The planned $q \in \{101, 151, 211\}$ tests are not merely robustness checks. They should determine whether the decay rate of $h_{\text{intra}}(\ell)$ is coherent across q , and whether it scales compatibly with the decay of the fibre-erasure functional $I(c; \sigma(\ell))$. If the two terms of decomposition (5) decay with compatible exponents, the decomposition becomes an operational structural statement rather than a purely formal identity.

6 Open Problems

[O-1] Analytical proof of [H-suff].

Establish that the BFS depth-increment map $G_{\ell+1 \leftarrow \ell}$ is a Markov kernel on σ : the transition $p(\sigma(\ell+1) \mid \sigma(\ell), c)$ is independent of c . Numerical evidence is available for $q \in \{61, 151, 211\}$ via the Spearman residual test of Section 5.3; the analytical proof is the remaining open task. The structural route would show that the BFS recursion depends on the Heisenberg group geometry — shared across all fibre labels up to conjugation — and the Born-Infeld admissibility constraint, not on the specific character of block c .

[O-1b] Proof of the semigroup hypothesis [H-sg].

Show that the coarse-graining channels compose as $G_{\ell_3 \leftarrow \ell_2} \circ G_{\ell_2 \leftarrow \ell_1} = G_{\ell_3 \leftarrow \ell_1}$ for all $\ell_1 < \ell_2 < \ell_3$. Without [H-sg], the data-processing inequality yields step-by-step monotonicity (from each application of [H-suff]) but not the uniform statement of Theorem 4.3 for all $\ell' > \ell$. Neither a numerical test nor an analytical proof is currently available.

[O-2] c -theorem analog via $H(\sigma(\ell))$.

The decomposition (5) identifies $H(\sigma(\ell) \mid c) = H(\sigma(\ell)) - I(c; \sigma(\ell))$ as a candidate counting monotone. Establish its monotonicity in ℓ and verify that its fixed-point value counts effective degrees of freedom. This would upgrade the fibre-erasure theorem to a genuine c -theorem analog. Preliminary numerical evidence for the monotonicity is available for $q = 61$ via the intra-block variance proxy (Section 5.5); analytical proof and cross- q confirmation ($q \in \{101, 151, 211\}$) remain open.

[O-3] Temporal causal structure and emergent light cone.

Define the temporal mutual information $I_{\text{temp}}(n, k) = I(U_n; U_{n+k})$ and the transfer entropy $T(U_n \rightarrow U_{n+k}) = I(U_{n+k}; U_n \mid U_{n-1}, \dots, U_0)$. Their decay in k would yield an operational definition of the emergent forward light cone on the temporal axis n , orthogonal to the present paper.

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